Name _____ Class _____

National 5 Engineering Science

Assignment: mark it yourself — worksheet



R Stewart - November 2026

Remember

This pack is for practice **before** your assignment. The real assignment is done on your own, with no help.

Do not copy answers from these scripts. Each year has a new task.

Use this worksheet with the **reader** on the computer. The reader has the task, the marking instructions, the three scripts and the answers.

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Grid 1: script 1

Script 1 is part 4 of the reader (17 pages). For each task, **write** your mark and what the marker would look for: is it here, or missing?

Task	What it asks for	Max	Your mark	What is here, or what is missing?
1a	Actuator and justification	2		
1b	Pneumatic circuit	5		
2a	Flowchart and circuit	5		
2b	High-level code	1		
2c	Test results and amendments	6		
2d	Amendments carried out	2		
3a(i)	System diagram: input and outputs	2		
3a(ii)	Sub-system diagram	6		
3b	Input sensing circuit	3		
3c	Test plan: Vout	3		
4a	Evaluate the gear design	3		
4b	Design a gear train	2		
4c	Simulate the gear train	2		
4d	Velocity ratio	2		
5a	Logic diagram	2		
5b	Simulate the logic	2		
5c	Modify and justify	2		
	Total	**50**		

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Grid 2: script 2

Script 2 is part 5 of the reader (16 pages). For each task, **write** your mark and what the marker would look for: is it here, or missing?

Task	What it asks for	Max	Your mark	What is here, or what is missing?
1a	Actuator and justification	2		
1b	Pneumatic circuit	5		
2a	Flowchart and circuit	5		
2b	High-level code	1		
2c	Test results and amendments	6		
2d	Amendments carried out	2		
3a(i)	System diagram: input and outputs	2		
3a(ii)	Sub-system diagram	6		
3b	Input sensing circuit	3		
3c	Test plan: Vout	3		
4a	Evaluate the gear design	3		
4b	Design a gear train	2		
4c	Simulate the gear train	2		
4d	Velocity ratio	2		
5a	Logic diagram	2		
5b	Simulate the logic	2		
5c	Modify and justify	2		
	Total	**50**		

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Grid 3: script 3

Script 3 is part 6 of the reader (16 pages). For each task, **write** your mark and what the marker would look for: is it here, or missing?

Task	What it asks for	Max	Your mark	What is here, or what is missing?
1a	Actuator and justification	2		
1b	Pneumatic circuit	5		
2a	Flowchart and circuit	5		
2b	High-level code	1		
2c	Test results and amendments	6		
2d	Amendments carried out	2		
3a(i)	System diagram: input and outputs	2		
3a(ii)	Sub-system diagram	6		
3b	Input sensing circuit	3		
3c	Test plan: Vout	3		
4a	Evaluate the gear design	3		
4b	Design a gear train	2		
4c	Simulate the gear train	2		
4d	Velocity ratio	2		
5a	Logic diagram	2		
5b	Simulate the logic	2		
5c	Modify and justify	2		
	Total	**50**		

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Focus tasks

The tasks are in order of marks lost, biggest first. Page numbers are the script page numbers in the reader.

Task 1: Test results and amendments (2c)

Why: Pupils in Scotland lost 3.0 of 6 marks here in 2024. This is the biggest loss in the whole assignment.

Look at: script 1 page 5; script 2 page 5; script 3 page 5.

1 **Tick** each thing each script gets right.

	Script 1	Script 2	Script 3
Test 1 result: LEDs flash and the motor does not turn			
Test 1 change: turn the diode round			
Test 1 re-test: the motor turns and the LEDs flash			
Test 2 result: LEDs off and the motor does not turn			
Test 2 change: loop back from output 7 off to the start			
Test 2 re-test: what the motor and LEDs do			

2 **Explain** why script 1's Test 1 result scored 0.

It scored 0 because

3 **Write** a Test 1 result that would gain the mark.

When the test switch is turned on,

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Task 2: The sub-system diagram (3a(ii))

Why: Pupils in Scotland lost 2.9 of 6 marks here in 2024. Each of the six marks needs one exact thing.

Look at: script 1 page 9; script 2 page 7; script 3 page 8.

1 Tick each thing each diagram has.

	Script 1	Script 2	Script 3
a system boundary, and a box round each sub-system			
the motor in an output position			
the lamp in an output position			
one driver for each output, each joined to the microcontroller			
two limit switches, each joined to the microcontroller			
a feedback loop from after the motor, through a limit switch			

2 Explain why script 3 did not gain the driver mark.

Script 3 did not gain it because

3 Describe where a feedback loop must start and finish to gain the mark.

The loop starts

and finishes

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Task 3: The pneumatic circuit (1b)

Why: Pupils in Scotland lost 1.4 of 5 marks here in 2024. No script in this pack gained every mark.

Look at: script 1 page 2; script 2 page 2; script 3 page 2.

1 Tick each thing each circuit has.

	Script 1	Script 2	Script 3
cylinder piped to the 5/2 valve			
two restrictors that slow both strokes			
both 3/2 valves piped to the 5/2 valve			
push button and actuator on the right 3/2 valves			
a reservoir after the instroking 3/2 valve			

2 Explain why no script gained the restrictor mark.

The mark needs

3 State where the reservoir goes, and which script left it out.

The reservoir goes

The script that left it out is

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Task 4: The test plan and Vout (3c)

Why: Pupils in Scotland lost 1.4 of 3 marks here in 2024. Every script in this pack scored 1 out of 3.

Look at: script 1 page 11; script 2 page 9; script 3 page 10.

1 **Tick** each thing each test plan does.

	Script 1	Script 2	Script 3
Test 1 result says what happens to Vout			
Test 2 is a suitable test			
Test 2 result says what happens to Vout			

2 **Explain** why script 1's expected results did not score.

They did not score because

3 **Write** both expected results so that they would gain the marks.

Test 1, reduce the light level:

Test 2, increase the light level:

Task 5: Modify and justify (5c)

Why: Pupils in Scotland lost 1.0 of 2 marks here in 2024. Half the marks were lost here.

Look at: script 1 page 17; script 2 page 16; script 3 page 16.

1 **Tick** each thing each answer gets right.

	Script 1	Script 2	Script 3
a valid change to the logic circuit			
a clear justification			

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2 Explain why script 3's justification gained the mark. "Harder to solve" on its own does not score.

It gained the mark because it says

3 Write a justification for adding another input to the circuit.

Adding another input

Task 6: System diagram, input and outputs (3a(i))

Why: Pupils in Scotland lost 1.0 of 2 marks here in 2024. Half the marks were lost here.
Look at: script 1 page 8; script 2 page 6; script 3 page 7.

1 Tick each thing each diagram gets right.

	Script 1	Script 2	Script 3
the input is a light level, not a component			
the outputs are light and movement, not components			

2 Explain why script 3 scored 0.

Script 3 scored 0 because

3 Write the input and the two outputs that gain both marks.

Input:

Outputs:

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Credits: candidate evidence 1 to 3 and commentaries, National 5 Engineering Science Assignment 2024, Understanding Standards, © Qualifications Scotland (SQA). Assessment task (toy shop) and finalised marking instructions, National 5 Engineering Science Assignment 2024, © Qualifications Scotland (SQA); photographs in the task: Task 1 Galchenkova Ludmila/Shutterstock.com, Task 5 Iuzvykova Iaroslava/Shutterstock.com. Marks lost: Centre Data 2024, all centres in Scotland, © Qualifications Scotland (SQA). Drawings by R Stewart. R Stewart - November 2026